Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	CASE-df048-apr-ios
Operator:	Floris
Analysis Started:	2026-06-15 15:37:34
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\train_mavic_air\df048\2018_april\ios
Output Directory:	C:\Users\Floris\Documents\dji_drones\train_mavic_air\df048\2018_april\dfdof_output\CASE-df048-apr-ios
Evidence Sources:	2
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner.The populated case output directory is the authorative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df048_internal.E01	drone_sd	physical	7.8 GB	2026-06-15 15:37:58
ios_logical.zip	controller_ios	logical	164.1 MB	2026-06-15 15:37:59

Evidence Source Path

C:\Users\Floris\Documents\dji_drones\train_mavic_air\df048\2018_april\ios

Cryptographic Hashes

df048_internal.E01		drone_sd	physical
SHA-256:	`5664f81579a3558ed7fdf79b06e49cf0849b5f41999ae5a996e43041460ac8f3`		
SHA-1:	`b9bfc2d90b326950aea2121be9304692b298a288`		

ios_logical.zip		controller_ios	logical
SHA-256:	`bfa6bf59fa10e999fc080b84ce13b108bd759e1d4bc9f2155caa48389070deea`		
SHA-1:	`94ab6d0ff0285bfe4de278ab8cc1a657aa10531d`		

2. Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p1 - drone sd]: mmls: no partition table found in df048_internal.E01; treating as direct partition image
2. [p2 - controller android]: source evidence not found
3. [p3 - controller android]: source evidence not found
4. [p3 - drone flight storage]: source evidence not found
5. [p4 - controller android]: source evidence not found
6. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
datcon	1	1	[OK]
exiftool	12	12	[OK]
fls	1	1	[OK]
icat	10	10	[OK]
mmls	1	0	[X] Failed
parser_ios	1	1	[OK]
txtlogtocsvtool	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	7 / 7
Flights With Primary Correlation	1 / 1
Tools Succeeded	6 / 7

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	com.dji.go	[K] Controller-only	1
Controller Device	iPad mini 4	[K] Controller-only	1
Drone Serial Number	0K1DECE2BC0633	[K] Controller-only	2
Drone Model	Mavic Air	[K] Controller-only	3
Controller Device	iPad5,1__weibosdk__003143000__iphone__os 10.3.2	[K] Controller-only	1
DJI Application Version	4.2.12	[K] Controller-only	2
Account E-Mail	Droneforensics@vtoinc.com	[K] Controller-only	1
Product Type	38	[K] Controller-only	1
Last Country Code	US	[K] Controller-only	1
Last App Launch	2018-04-19T17:24:45Z	[K] Controller-only	1
Last Flight Date	1970-01-01T00:00:00Z	[K] Controller-only	1
Last Known Latitude (Find Aircraft)	39.96127315	[K] Controller-only	1
Last Known Longitude (Find Aircraft)	-106.21652817	[K] Controller-only	1
Last Known Altitude (Find Aircraft)	5.7	[K] Controller-only	1
Last Known Heading (Find Aircraft)	-125.1	[K] Controller-only	1
Last Known Horiz. Accuracy (Find Aircraft)	1.0	[K] Controller-only	1

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status:	Matched (primary GPS+temporal)
Confidence:	medium
GPS Overlap:	160.5 s
Median GPS Distance:	14.13 m
Overlap:	100%
GPS Match Rate:	100%
Flight Record SHA-256:	`90de266a6242dea7da95bf08d9592c5357b69b0f91cb96f01454e090ed5e4dd7`
Flight Record Name:	DJIFlightRecord_2018-04-19__11-25-10_.csv
Start:	2018-04-19T17:25:11.078000+00:00
End:	2018-04-19T17:27:51.546000+00:00
Duration:	160.5 s
Drone Log SHA-256:	`ae079e268e4e114016b05125037569fd076c2feb783e54611c287ca51f13300f`
Drone Log Name:	2018-04-19_11-24-49_FLY029.csv
Start:	2018-04-19T17:15:11.459000+00:00
End:	2018-04-19T17:28:12.470000+00:00
Duration:	781.0 s

Flight Metrics

Peak Height:	56.8 m
Distance Travelled:	2013.46 m
Video Recording:	Yes
Photo Capture:	No
Photos Taken:	0
Recording Duration:	113.0 s
Record Complete:	Yes

Fly Modes Observed (Chronological)

1. GPS_Atti
2. ASST_TAKEOFF
3. EngineStart
4. AssitedTakeoff
5. AssistedTakeoff
6. SPORT
7. Sport
8. FORCE_LANDING
9. ForceLanding
10. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2018-04-19 17:24:49	ctrl_ios:dl	Log started	MavicAir
2018-04-19 17:24:49	ctrl_ios:dl	Motor turned off	off
2018-04-19 17:25:10	ctrl_ios:dl	Motor turned on	on
2018-04-19 17:25:11	ctrl_ios:fr (v4.2.12)	Log started	SN: 0K1DECE2BC0633; App: 4.2.12; Mavic Air
2018-04-19 17:25:11	ctrl_ios:fr (v4.2.12)	Motor turned on	on
2018-04-19 17:25:11	ctrl_ios:fr (v4.2.12)	Record mode changed	Starting
2018-04-19 17:26:23	ctrl_ios:fr (v4.2.12)	Reached peak height	56.8 m
2018-04-19 17:26:25	ctrl_ios:dl	Reached peak height	54.7 m
2018-04-19 17:27:03	ctrl_ios:fr (v4.2.12)	Record mode changed	Stop
2018-04-19 17:27:04	ctrl_ios:fr (v4.2.12)	Record mode changed	No

Timestamp (UTC)	Source	Event	Details
2018-04-19 17:27:51	ctrl_ios:dl	Motor turned off	off
2018-04-19 17:27:51	ctrl_ios:fr (v4.2.12)	Log ended	Dist home: 1.3 m; Height: 0.0 m; Rec: 113.0 s
2018-04-19 17:27:51	ctrl_ios:fr (v4.2.12)	Motor turned off	off
2018-04-19 17:28:13	ctrl_ios:dl	Log ended	

In-Flight Warnings

Timestamp (UTC)	Source	Warning Message
2018-04-19 17:26:34	ctrl_ios:fr (v4.2.12)	Warning:In Flight. working compass encounters magnetic-field interference.please switch to atti mode if craft behave abnormally
2018-04-19 17:26:34	ctrl_ios:fr (v4.2.12)	Warning:In Flight. working compass encounters magnetic-field interference.please switch to atti mode if craft behave abnormally
2018-04-19 17:26:42	ctrl_ios:fr (v4.2.12)	Warning:In Flight. working compass encounters magnetic-field interference.please switch to atti mode if craft behave abnormally
2018-04-19 17:26:44	ctrl_ios:fr (v4.2.12)	Warning:In Flight. working compass encounters magnetic-field interference.please switch to atti mode if craft behave abnormally
2018-04-19 17:26:55	ctrl_ios:fr (v4.2.12)	Warning:In Flight. working compass encounters magnetic-field interference.please switch to atti mode if craft behave abnormally
2018-04-19 17:26:55	ctrl_ios:fr (v4.2.12)	Warning:In Flight. working compass encounters magnetic-field interference.please switch to atti mode if craft behave abnormally
2018-04-19 17:27:00	ctrl_ios:fr (v4.2.12)	Warning:In Flight. working compass encounters magnetic-field interference.please switch to atti mode if craft behave abnormally

Correlated Artefacts

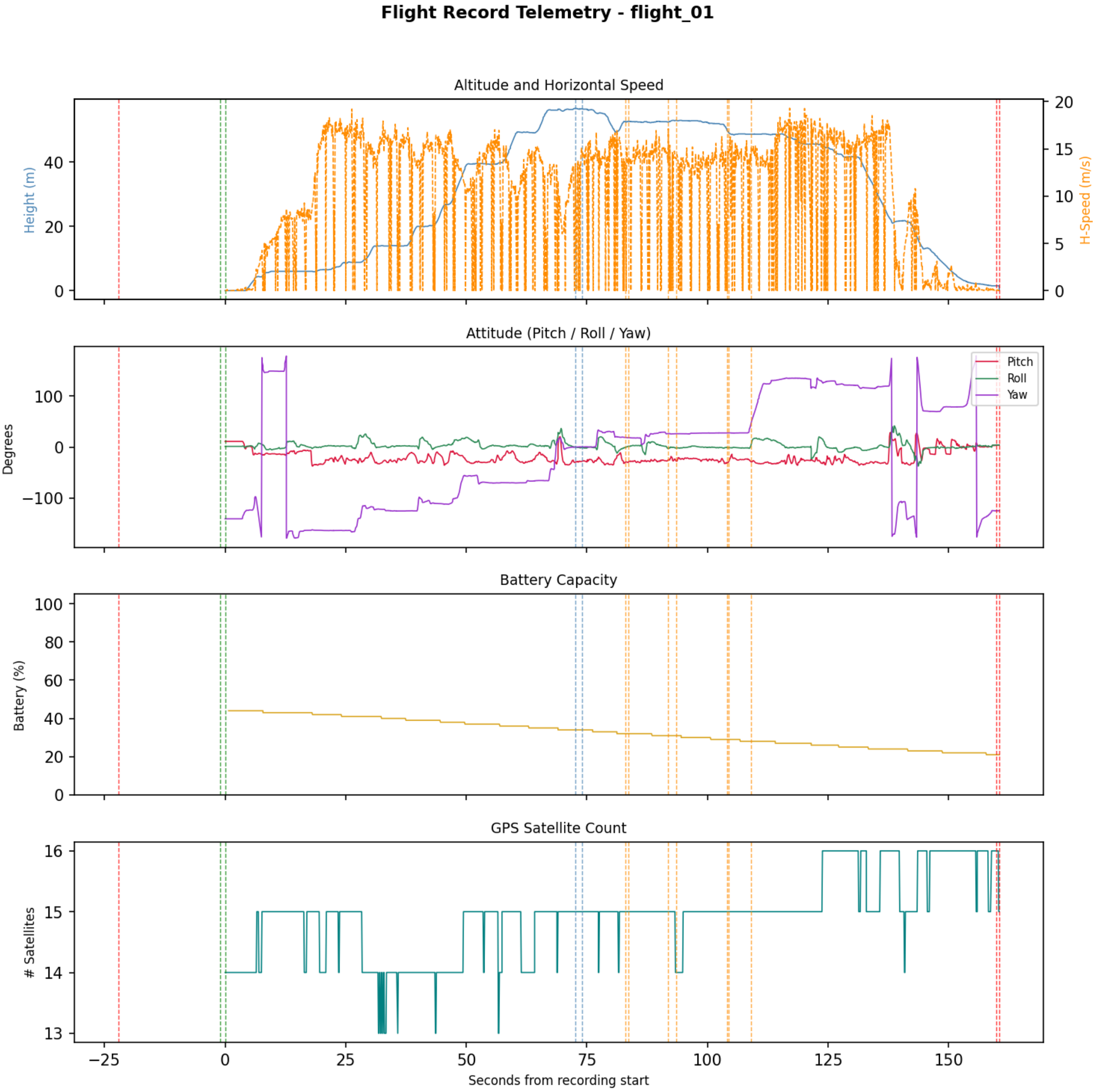
Possibly correlated (indirect indicators):

controller_ios: flight_logs	Log messages match (9 entries, format: json_array_log)
p5:1131c8b1540caabb232d177c1e94cf0411c16d77d5cc3dd3e82194dd3e6351de	

controller_ios:videos	Video duration 113.234 s matches log recording 113.0 s
p4:a50b36359ed71de2fa8558c88a9ddc27ccb0b334f3f6423f194f604762e6679d	

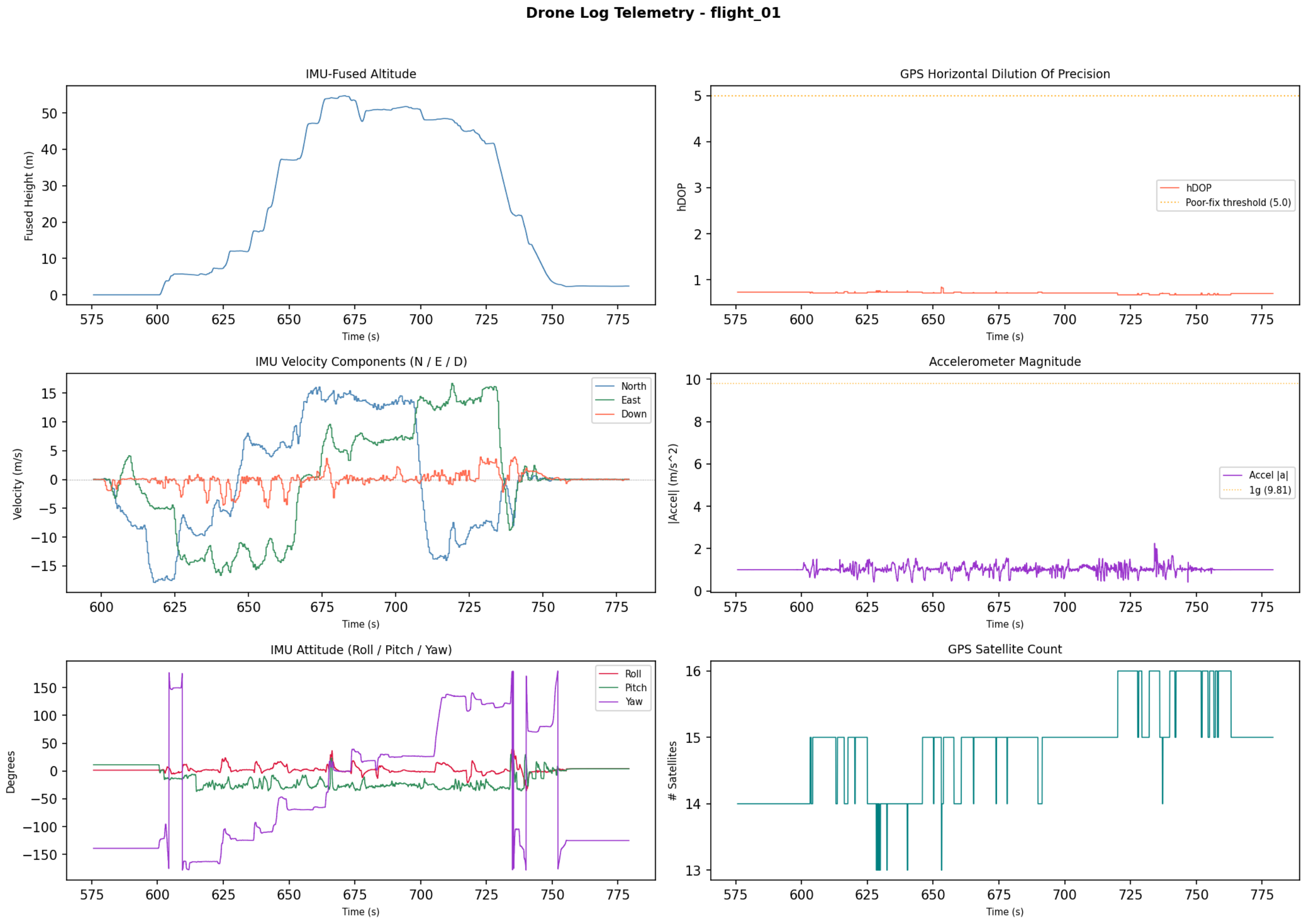
drone_sd:videos	Video duration 114.081 s matches log recording 113.0 s
p4:890e7c07f2d747cdbdffe8939636fc1c4f1230ce592db8d91fff8742551f8192	

Flight Record - Sensor Telemetry



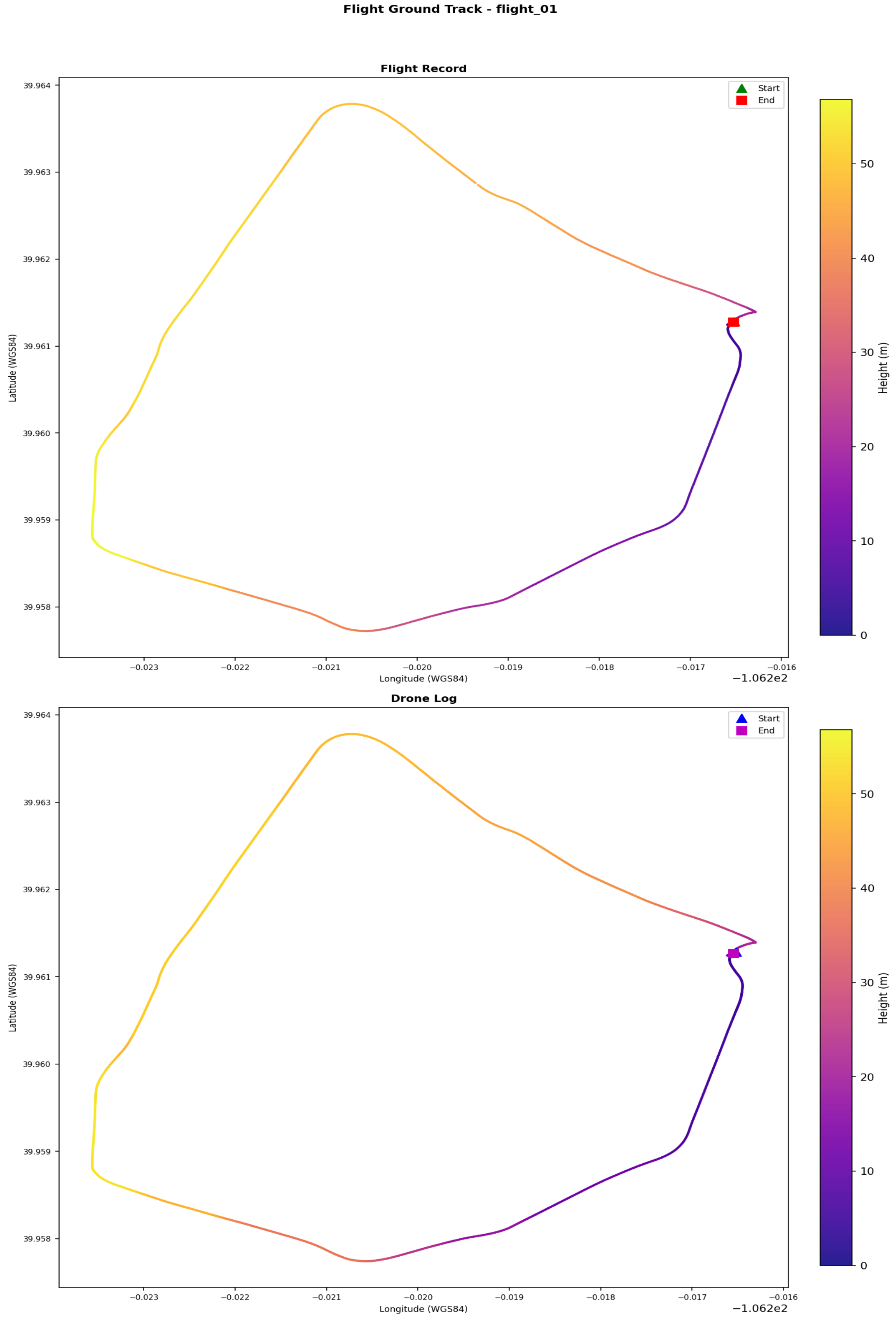
GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: 2018-04-19_11-24-49_FLY029.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (2018-04-19_11-24-49_FLY029.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path.

Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	No
controller_ios	Yes
drone_sd	Yes
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_ios	1	8	1	3	1	1	1
drone_sd	0	0	0	0	0	5	5

Data Quality Notes per Category

Databases (8 files):

- 4 database(s) empty (controller_ios)

Drone Logs (1 files):

- 1 file(s) with row anomalies
- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

Flight Logs (3 files):

- formats: 2 bracketed_log, 1 json_array_log

Flight Records (1 files):

- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

Images (6 files):

- 6 file(s) missing (normalisable) EXIF timestamp
- 6 file(s) missing EXIF GPS coordinates

Videos (6 files):

- 6 file(s) missing (normalisable) EXIF timestamp
- 6 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Drone_logs: 1 file(s) with row anomalies detected in telemetry data.
- Drone_logs: 1 file(s) with empty columns - some telemetry channels contain no data.
- Drone_logs: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.
- Flight_logs: log files present in the following format(s): 2 bracketed_log, 1 json_array_log.
- Flight_records: 1 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p1 - drone sd]: mmls: no partition table found in df048_internal.E01; treating as direct partition image
2. [p2 - controller android]: source evidence not found
3. [p3 - controller android]: source evidence not found
4. [p3 - drone flight storage]: source evidence not found
5. [p4 - controller android]: source evidence not found
6. [p4 - drone flight storage]: source evidence not found

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Databases

collectionList.db

- database_empty: all tables empty

djiFMDB.db

- database_empty: all tables empty

flysafe_app_dynamic_areas.db

- database_empty: all tables empty

flysafe_license.db

- database_empty: all tables empty

Drone Logs

norm_2018-04-19_11-24-49_FLY029.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 4 row(s) affected
- motor_airborne_off: 338 row(s) affected
- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATTI(0):absoluteHeight:C, IMU_ATTI(0):distanceHP:C, IMU_ATTI(1):absoluteHeight:C (+1 more)
- contains_constant_value: 48 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:BatAuth:D, BatteryInfo:FullCap:D (+43 more)

Flight Logs

2018-04-19_11_25_15-0K1DECE2BC0633.dat

- format: json_array_log | entries: 9

upgradeLog_0.txt

- format: bracketed_log | entries: 14

upgradeLog_1.txt

- format: bracketed_log | entries: 2

Flight Records

norm_DJIFlightRecord_2018-04-19__11-25-10_.csv

- contains_no_value: 40 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_SER_WARN.warn, CENTER_BATTERY.connStatus (+35 more)

- contains_constant_value: 165 column(s) - APP_TIP.tip, CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.enabledPhoto, CAMERA_INFO.encryptStatus (+160 more)

Images

2018_04_19_11_25_09.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

Videos

2018_04_19_11_25_09.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no firmware version found - Device and account information.
2. 4 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

com.dji.go.plist	Account Data	controller_ios	extract_logical
SHA-256:	2ccaf116dc71cbccda1612b0103e10b6a51d1f9321505edf29bceaa67e40acf3		
SHA-1:	382988b0f4531e1d61b4f5998527161b59fc2791		

dji_force_update.db	Databases	controller_ios	extract_logical
SHA-256:	6fac7adb4742381e578ecff38af1e2fa82eec71222fc9d6ef18e9c79aa799a46		
SHA-1:	105be3f68ad143cb7ad40cc0edfd23455ecb2b0b		

flysafe_license.db	Databases	controller_ios	extract_logical
SHA-256:	ef071ebc84923187a5c8819ddf39b58a160aaebd0e8f75a7e41a0fcc90c9da33		
SHA-1:	037c6f40d49371285301094fd840cfde0ee41633		

flysafe_app_dynamic_areas.db	Databases	controller_ios	extract_logical
SHA-256:	7ca264ba8538a4ed920c02222c3898714c0a8199da67bcf029b0e9ed2a36f6ab		
SHA-1:	759c3d9368daf2a307dcc53729c6eccc40fcab66		

flysafe_areas_djigo.db	Databases	controller_ios	extract_logical
SHA-256:	d9164be112208833fe065839204595086757 42a54e3ca70d358789ab362c7d3c		
SHA-1:	936e7b4a02e38dc7cd2b2d4cb7be8280096b17bd		

flysafe_dji_flight_dynamic_areas.db	Databases	controller_ios	extract_logical
SHA-256:	3aaaa2157a9465749e0460f1438b0729840e7a06309a596b5f07afee2947c4dc		
SHA-1:	14ff56247dd37de7754bf6af526aa56da1ba2512		

flysafe_static_circle.db	Databases	controller_ios	extract_logical
SHA-256:	4787fc859c865dab6b3d0db8eb68a55c764ca5d13105b14429cc2d455708cfb4		
SHA-1:	ccf2d5995e798abe1f1255c252c28ec8ae699dd4		

collectionList.db	Databases	controller_ios	extract_logical
SHA-256:	e8eb3b33022bd1e425301a6629eb4fd539409851892c02fc394b6f1092de54b6		
SHA-1:	5be96d0c777ce86a89d237e11bae0e0358b337af		

djiFMDB.db	Databases	controller_ios	extract_logical
SHA-256:	c2f956359b0006c3cf0a1ecad392d8c70da65f2ef9145fa6b995b84653329638		
SHA-1:	9d77797e6788fe25f2d71c838c7d59eae0164680		

2018-04-19_11-24-49_FLY029.DAT	Drone Logs	controller_ios	extract_logical
SHA-256:	68be26ad31842e7e345b3bb0f957cc8990b8a05625faceea26f7636fb6904041		
SHA-1:	cbf596350926fb5afc76249964640d8958bcda07		

2018-04-19_11_25_15-0K1DECE2BC0633.dat	Flight Logs	controller_ios	extract_logical
SHA-256:	1131c8b1540caabb232d177c1e94cf0411c16d77d5cc3dd3e82194dd3e6351de		
SHA-1:	89f5cf51110f874c17ca168f6879c381f78f1357		

upgradeLog_0.txt	Flight Logs	controller_ios	extract_logical
SHA-256:	c9932a9fca834fa338ede7545e0e437c4977ed3aeb5ee4304f04623a8a8a788f		
SHA-1:	fb6e51a24b9bd88973b13b9dbbdb7075ea6a8a49		

upgradeLog_1.txt	Flight Logs	controller_ios	extract_logical
SHA-256:	0b2010163a8b634a1087d572e910f6aa6080a2ad88a8ac8f461419e4b57eb175		
SHA-1:	232d159cd7cc53e37bd308113a04c56f312fa83a		

DJIFlightRecord_2018-04-19__11-25-10_.txt	Flight Records	controller_ios	extract_logical
SHA-256:	70665f6b87164fef6f9c21bdb96ef7a9a6709b337900af71650d4f76ae1a2d82		
SHA-1:	e3d3a12f2b031c77344ee78b0cbf44a13013a058		

2018_04_19_11_25_09.thumbnail	Images	controller_ios	extract_logical
SHA-256:	cc0db7cb2c455432bc5840dd637bf089200b6bd1c1c2031cdb47fa2a2b1b1fd3		
SHA-1:	fb34a607c16e985e6a0c0a02d2476eddbcee36f1		

2018_04_19_11_25_09.mp4	Videos	controller_ios	extract_logical
SHA-256:	a50b36359ed71de2fa8558c88a9ddc27ccb0b334f3f6423f194f604762e6679d		
SHA-1:	d0d00f3fbef6b9e4464a000b90c315d2dcce2c1f		

DJI_0001.THM	Images	drone_sd	extract_physical
SHA-256:	3a3918e0c00dddb9ad706491020da1b7db90479f49e29814274087259da2bcf0		
SHA-1:	b6b5af5e4d4b0d5ecae0eef6e7901ec873ce8048		

DJI_0002.THM	Images	drone_sd	extract_physical
SHA-256:	c452f70bc9e9bad9f05047d07cf685498ab5117ec940a5bc6622bd77baa79b1f		
SHA-1:	9f15bf98c1c26619f2499ece191f0dc426aae6e3		

DJI_0003.THM	Images	drone_sd	extract_physical
SHA-256:	9727e9310aa901a379b5fdfaf29a50e22fb879ff86d7ba6d75e82c6a39497d79		
SHA-1:	c958de01cff11f28d30a028a327b0d9867c968fc		

DJI_0004.THM	Images	drone_sd	extract_physical
SHA-256:	f99c4b6f1f614bd2ea9829aa50e08d966ed188dde73ce9dfcf388ee70adb7dc4		
SHA-1:	434ae634442c15bfbe0b51dbc2b31193f14968d5		

DJI_0005.THM	Images	drone_sd	extract_physical
SHA-256:	b3f920d7ad5412a19b8f122aaa0ff4cedfcf9b25c2baa89bbf4d26e0d796fe5f		
SHA-1:	6f4b46f652887ad89b44020dc176a4ac9f0bb324		

DJI_0001.MP4	Videos	drone_sd	extract_physical
SHA-256:	61a290fa102ce3e2913242b83aae8073af3751cd6b8f1ed0e14eab7625f77815		
SHA-1:	9873e6568f8631e23ae819daa214f6e88312efea		

DJI_0002.MP4	Videos	drone_sd	extract_physical
SHA-256:	9db5019a6c22e8ce2c2d726df4f8c1c0b52fc3d5806b38058bf1896f80e6ba7b		
SHA-1:	5131359773d614150ff1d62d8ee4c4d984eb640f		

DJI_0003.MP4	Videos	drone_sd	extract_physical
SHA-256:	aad026be3d81c7dbf251c37d7be4776d239dd44cd7a9988190e6a0f04b87796a		
SHA-1:	e74bf86c588742fb6cf85fda3b641c41e7834b65		

DJI_0004.MP4	Videos	drone_sd	extract_physical
SHA-256:	6aa7dc8ae4f82d9238d99673a3b4c30ab87f3dfe46627de30b2d6f200beaacb3		
SHA-1:	8a6bfb4eca3908e47a7bab171aa907e333ed0ab7		

DJI_0005.MP4	Videos	drone_sd	extract_physical
SHA-256:	890e7c07f2d747cdbdffe8939636fc1c4f1230ce592db8d91fff8742551f8192		
SHA-1:	a262dd9af22576b92269280cc9e1fe567e375821		

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Ios

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	1	1	0
Databases	8	0	0
Drone Logs	1	2	1
Flight Logs	3	0	0
Flight Records	1	1	1
Images	1	1	1
Videos	1	1	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	5	5	5
Videos	5	5	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-15 15:37:34 mmls vThe Sleuth Kit ver 4.15.0 rc=1

cmd: mmls.exe df048_internal.E01

2026-06-15 15:37:34 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p df048_internal.E01

2026-06-15 15:38:16 parser_ios v- rc=n/a

cmd: ios_logical.zip C:\Users\Floris\Documents\dfdof_output\CASE-df048-apr-ios\p2_image_parsing\controller_ios_parsed

out: controller_ios_parsed

2026-06-15 15:38:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1357829

out: DJI_0001.MP4

2026-06-15 15:38:24 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1357830

out: DJI_0002.MP4

2026-06-15 15:38:37 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1357831

out: DJI_0003.MP4

2026-06-15 15:38:53 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1357832

out: DJI_0004.MP4

2026-06-15 15:39:04 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1357833

out: DJI_0005.MP4

2026-06-15 15:39:09 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1358853

out: DJI_0001.THM

2026-06-15 15:39:09 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1358854

out: DJI_0002.THM

2026-06-15 15:39:09 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1358855

out: DJI_0003.THM

2026-06-15 15:39:09 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1358856

out: DJI_0004.THM

2026-06-15 15:39:09 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1358857

out: DJI_0005.THM

2026-06-15 15:39:34 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe 2018-04-19_11-24-49_FLY029.DAT

out: 2018-04-19_11-24-49_FLY029.csv, 2018-04-19_11-24-49_FLY029.kml

2026-06-15 15:39:35 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2018-04-19__11-25-10_.txt DJIFlightRecord_2018-04-19__11-25-10_.csv

out: DJIFlightRecord_2018-04-19__11-25-10_.csv

2026-06-15 15:39:35 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2018_04_19_11_25_09.thumbnail

out: 2018_04_19_11_25_09_exif.json

2026-06-15 15:39:36 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2018_04_19_11_25_09.mp4

out: 2018_04_19_11_25_09_exif.json

2026-06-15 15:39:36 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM

out: DJI_0001_exif.json

2026-06-15 15:39:36 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.THM

out: DJI_0002_exif.json

2026-06-15 15:39:37 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.THM

out: DJI_0003_exif.json

2026-06-15 15:39:37 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.THM

out: DJI_0004_exif.json

2026-06-15 15:39:37 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.THM

out: DJI_0005_exif.json

2026-06-15 15:39:38 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MP4

out: DJI_0001_exif.json

2026-06-15 15:39:38 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.MP4

out: DJI_0002_exif.json

2026-06-15 15:39:38 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.MP4

out: DJI_0003_exif.json

2026-06-15 15:39:39 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.MP4

out: DJI_0004_exif.json

2026-06-15 15:39:39 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.MP4

out: DJI_0005_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap >= 60 s AND >= 90% of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs <= 25 m with >= 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a +-2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.